Programming assignment: Discrete-event simulation of a queue

Due: Noon on Monday, March 3 2025 on Moodle
Maximum marks: 15

Your task is to implement a discrete event simulator, preferably in Python, to simulate packet arrivals and departures from a queue. Each packet should have a distinct integer sequence number starting at sequence number 0. Choose an appropriate data structure to hold simulation events.

The length of a simulation run is a free parameter that you can choose. Given this length, for each simulation run r, compute the mean waiting time in the queue w_r. You should run as many simulations as necessary so that the confidence interval of w (computed over multiple simulation runs) is less than 5% of its mean w_{ave}.

Simulate both an M/M/1 and an M/D/1 queue. Use appropriate statistical libraries as needed.

Turn in the code as a Jupyter notebook on Moodle. Please also use the notebook to answer the following questions.

1. List the event types in your simulation. Why do you need each of these events? (1 mark)
2. What were the fields you used to describe each event? Why? (1 mark)
3. What data structure did you use to hold simulation events? Why did you choose this data structure? (1 mark)
4. For the M/M/1 simulation, show a trace of 10 packet arrivals and their departures in the format:

   <timestamp>: the current queue length, event type, event details // comment

   for the case where utilisation = 0.8.
   Use the comments to explain what is happening in each step of the simulation. (3 marks)
5. You will find that you can achieve the stopping criterion either by choosing a longer simulation length and performing fewer runs or choosing a shorter simulation length and needing to perform more simulation runs. Which approach takes less computation time? Why? Can you suggest a way to find the optimal duration of a simulation run? (2 marks)
6. When the simulation run duration is too short, even when the stopping criterion is met, a problem arises. What is it? Why does it happen? What cautionary lesson can one derive from this? (3 marks)
7. Once you have chosen the simulation run length, compare the simulation results with the corresponding theoretical values by having your code draw a graph for each queue type (M/M/1 and M/D/1). In each graph, the X axis should be the utilisation (rho) and the Y axis plots both
   a. the mean waiting time in the queue computed by the simulation
   b. the mean waiting in the queue that is computed using the appropriate formula. For the simulation results, also show the confidence interval. (4 marks)

Note: Tools such as ChatGPT and Co-Pilot will let you automatically generate big chunks of the code. However, they will not improve your understanding of discrete event simulation or queues. For this reason, please don't use these tools.